Introduction to Computer Networks

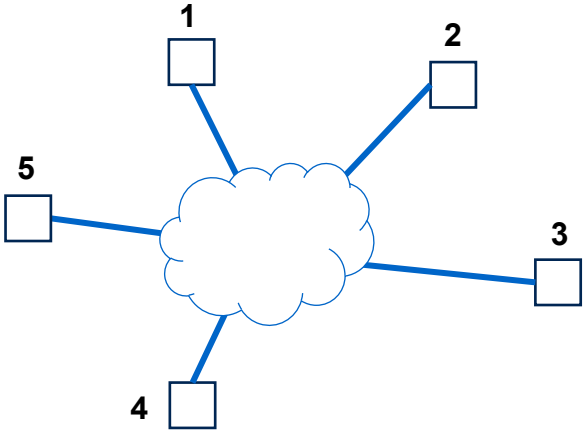
IT5100D AY25/26 S2

NUS | Computing
National University of Singapore

© Copyright National University of Singapore. All Rights Reserved.

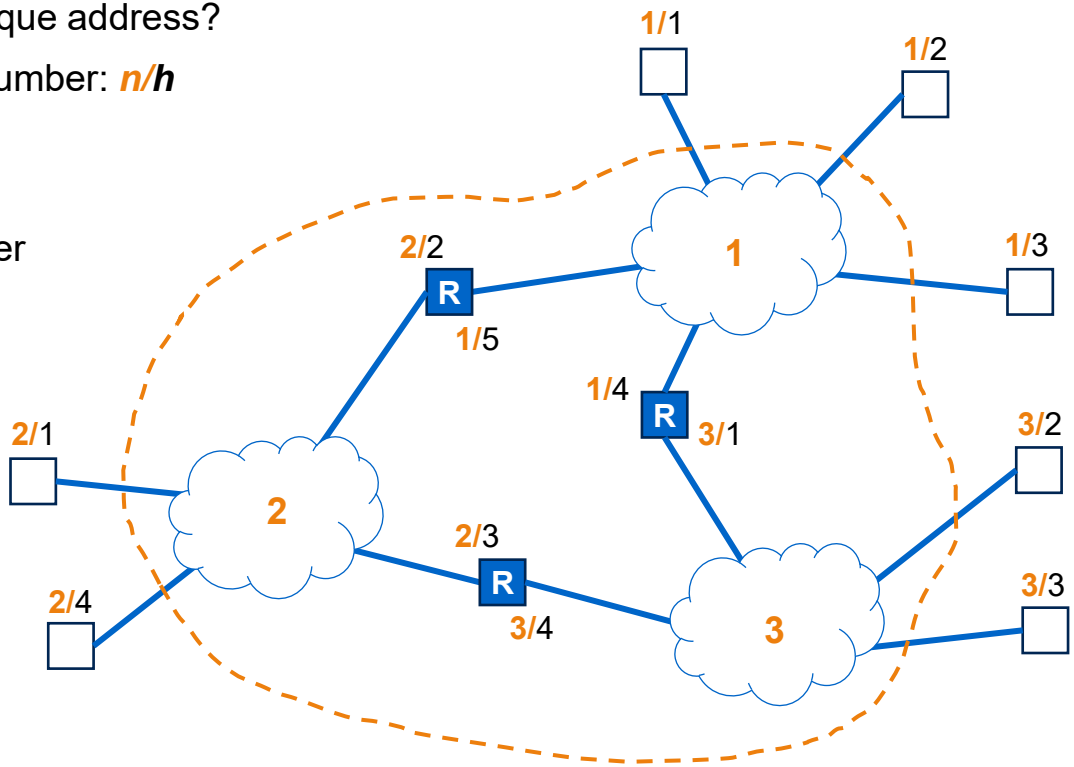
What is a Network?

- A *network*: A group of devices that can communicate with each other over *links*.
- Each device is called a *host* or *end system*.
- Each host has a unique *address*.



What is an Internet?

- An *internet*: a network of networks (an ***interconnected network)***
- How can each host have a *globally* unique address?
- Simplistically, prefix with the network number: ***n/h***
  - ***n***: the network number
  - ***h***: the number of the host on network *n*
- We will learn the actual mechanism later



R Router

The Internet with a Capital “I”

“The words *internetwork* and *internet* are simply a contraction of the phrase *interconnected network*.

However, when written with a capital “I”, the Internet refers to the worldwide set of interconnected networks.

Therefore, the Internet is an internet, but the reverse does not apply.”

-- From: IBM, “TCP/IP Tutorial and Technical Overview” (8th ed., 2006)

Also:

The Internet – the global computer network (the infrastructure)

vs

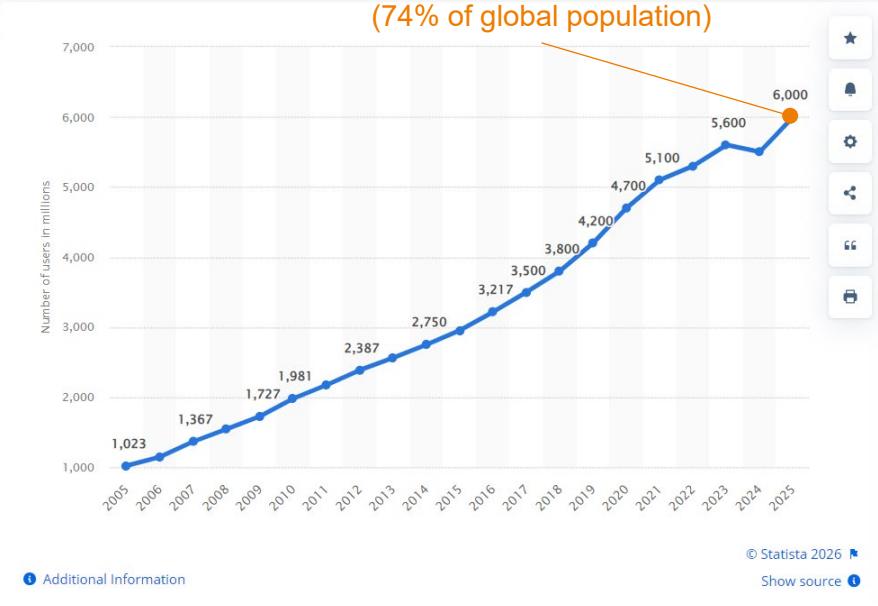
World Wide Web (“the Web”, WWW) – the “distributed hypertext system” [Berners-Lee, 1989] most commonly used to “browse” resources on the Internet

How Big is the Internet?

Internet › Demographics & Use

Number of internet users worldwide from 2005 to 2025

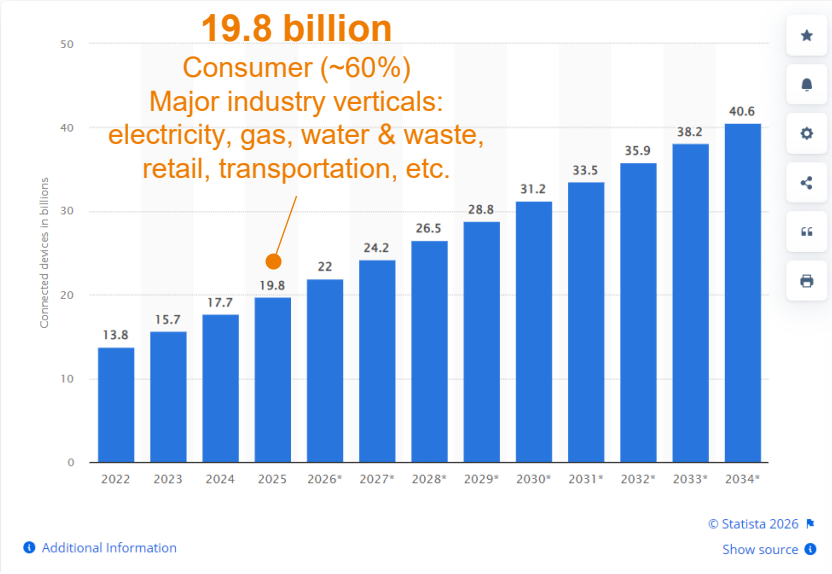
(in millions)



Technology & Telecommunications › Telecommunications

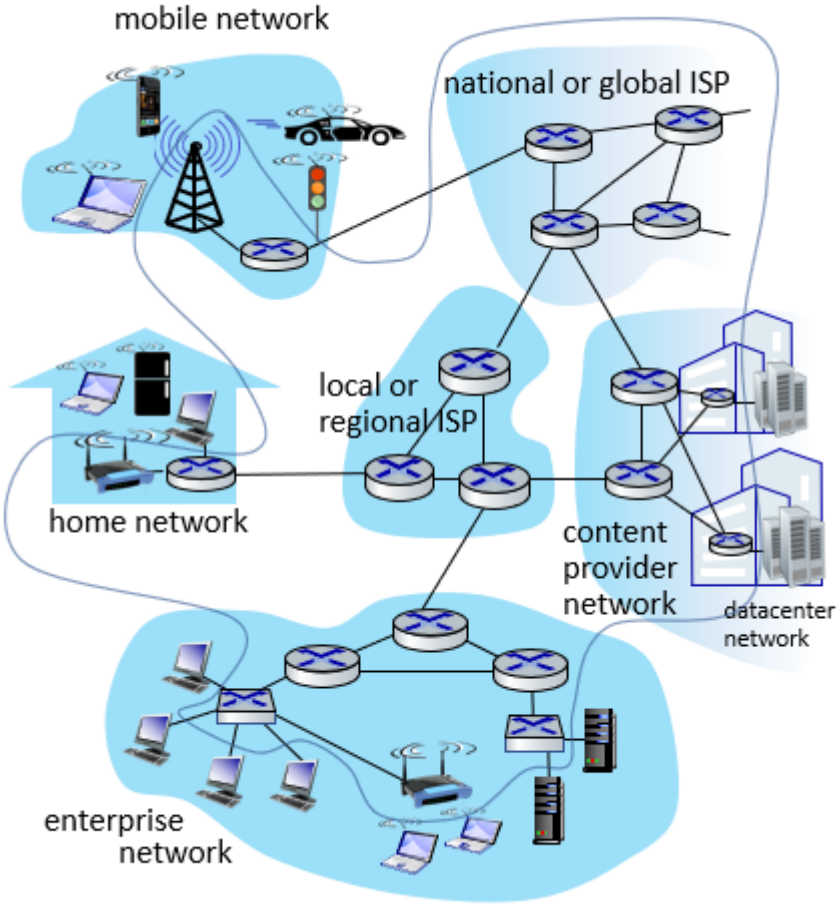
Number of Internet of Things (IoT) connections worldwide from 2022 to 2023, with forecasts from 2024 to 2034

(in billions)



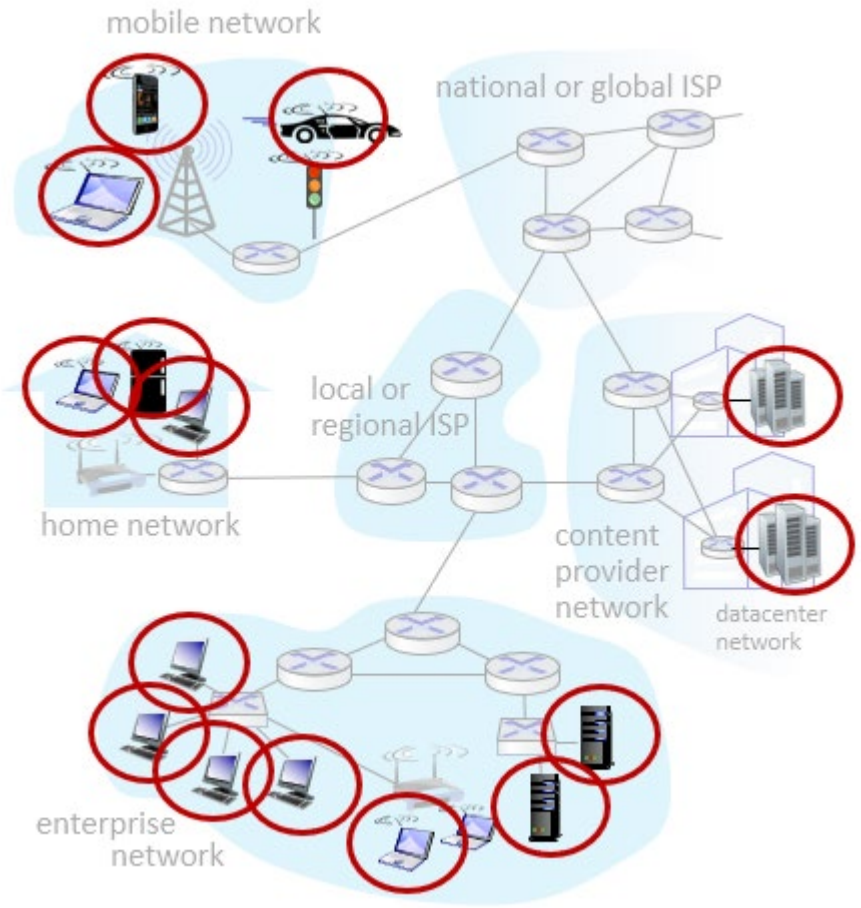
Internet Structure

Let's use this illustration to learn the terminology.



Network Edge

- Hosts (also called *end systems*)

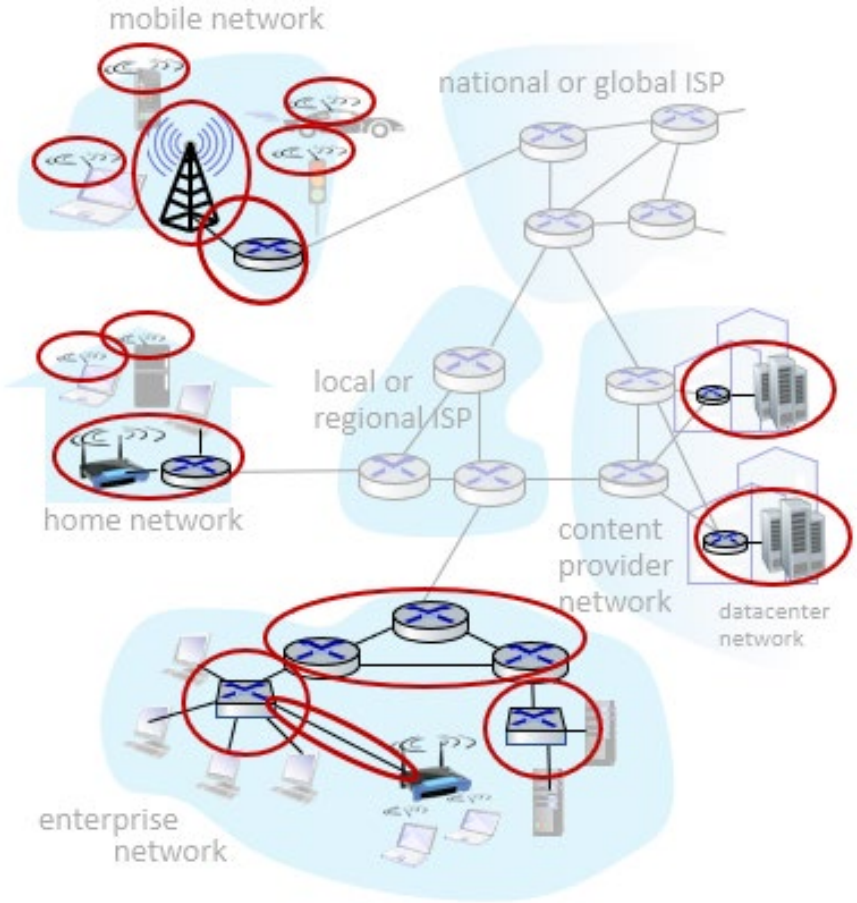


Network Edge

- Hosts (also called *end systems*)
- *Access networks*: communication links that connect the hosts to their *immediate service provider*
  - A *Local Area Network (LAN)* connects end systems to the *edge router*

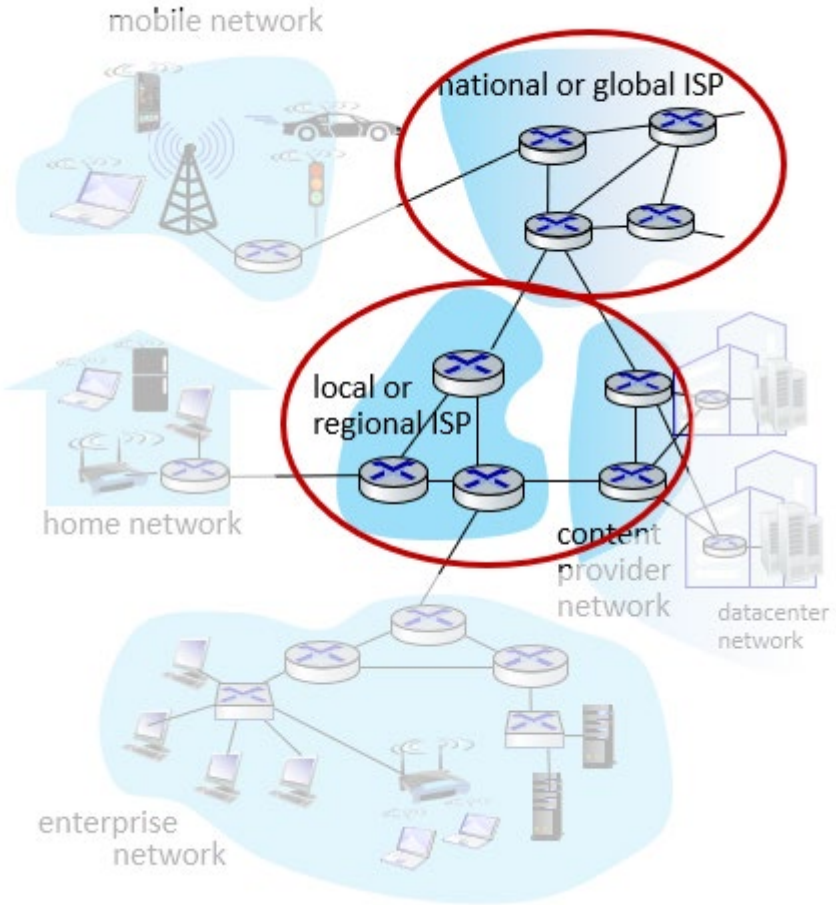
Switch → connects end systems together to create a LAN

Router → connects switches and their respective networks together and allows them to access the Internet



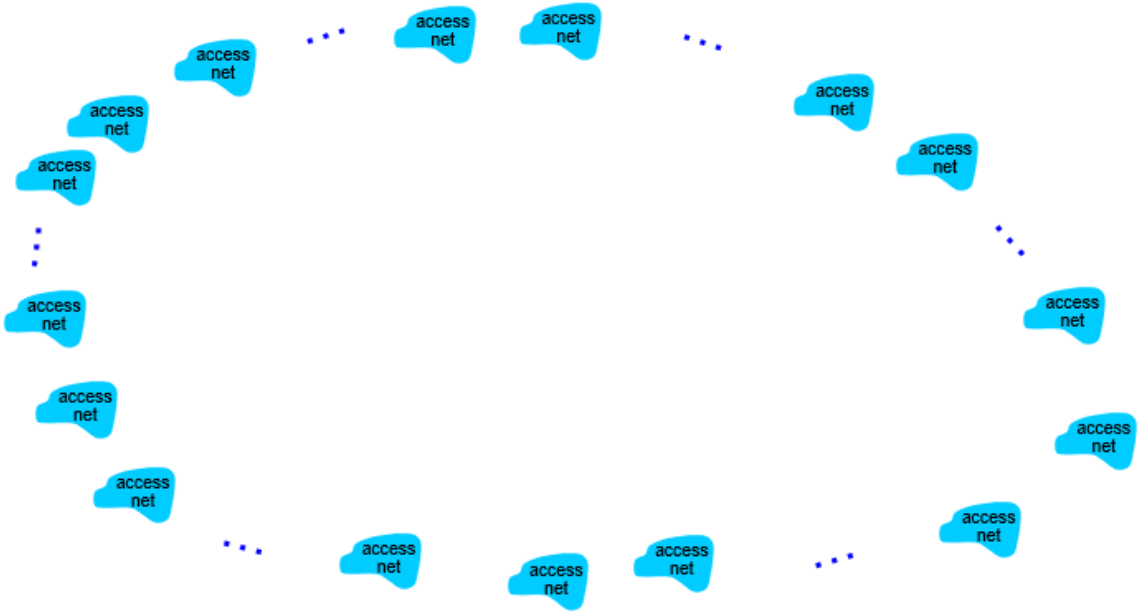
Network Core

- Interconnected routers
- Network of networks
- **ISP:** Internet Service Provider
  - Connect hosts to the Internet
  - Must be interconnected!
  - Hierarchically organized



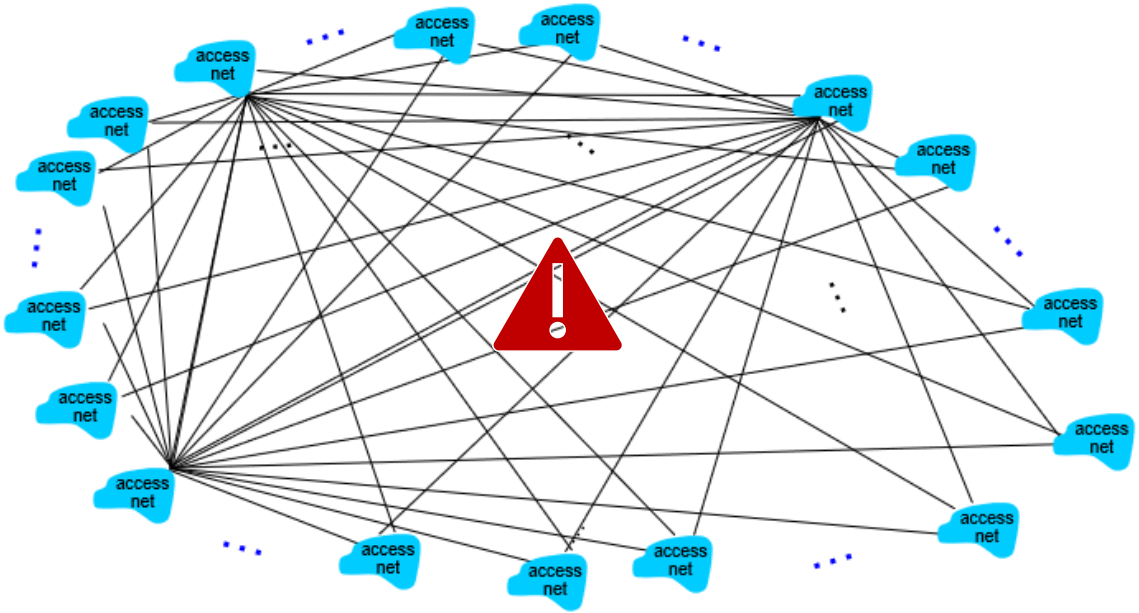
Internet Structure: A Network of Networks

- Let's take a stepwise approach to describe the current Internet structure
- Given <u>millions</u> of access ISPs, how to connect them together?



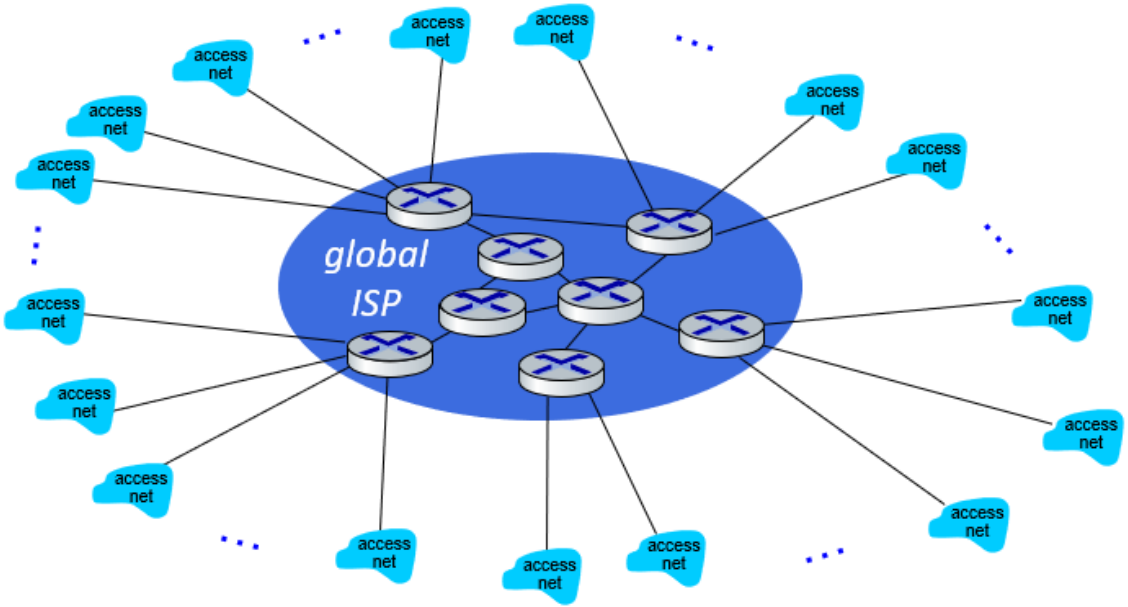
Internet Structure: A Network of Networks

- Connect each ISP to each other directly?
- $O(N^2)$ connections → **does not scale**



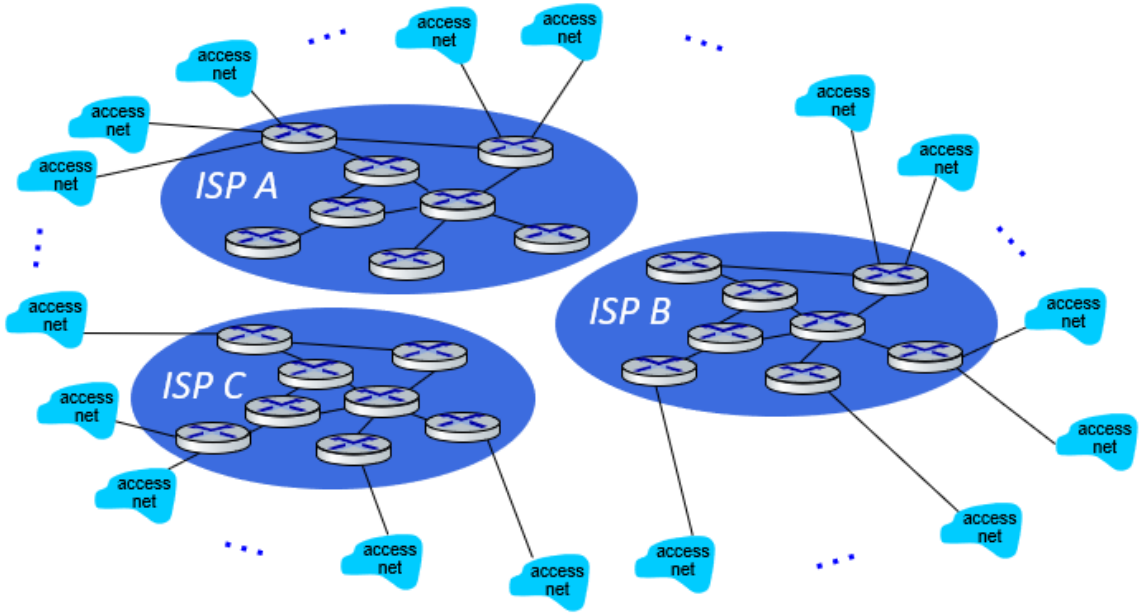
Internet Structure: A Network of Networks

- Option: connect each access ISP to one *global transit ISP*
- *Customer* and *provider* ISPs have economic agreement



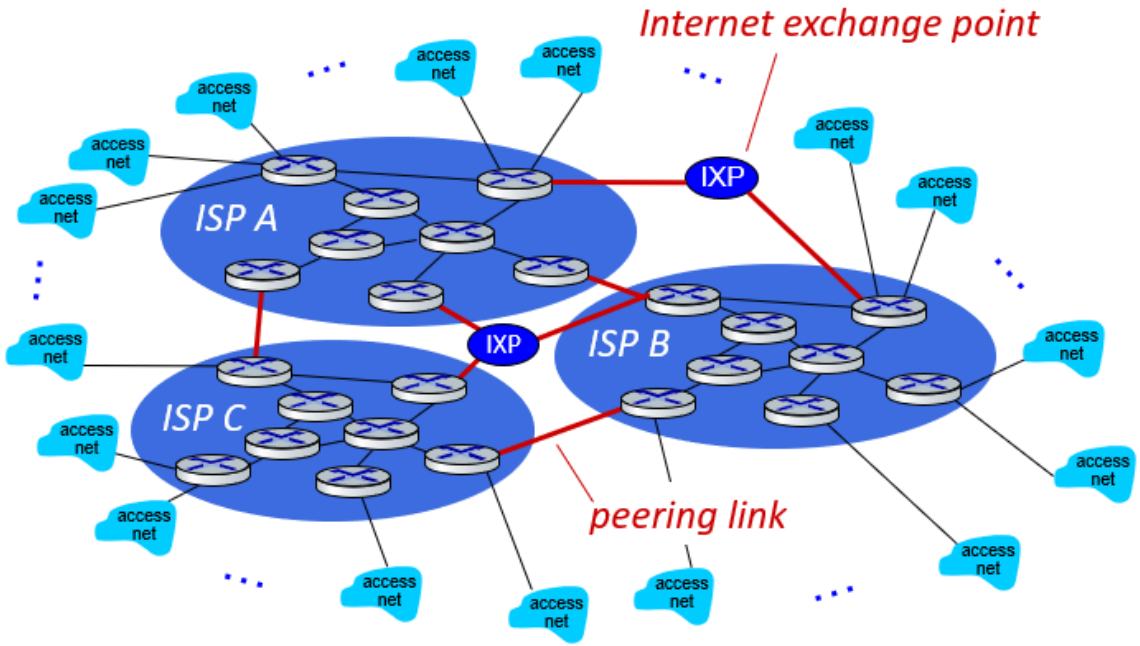
Internet Structure: A Network of Networks

- But if one global ISP is viable business, there will be *competitors*…



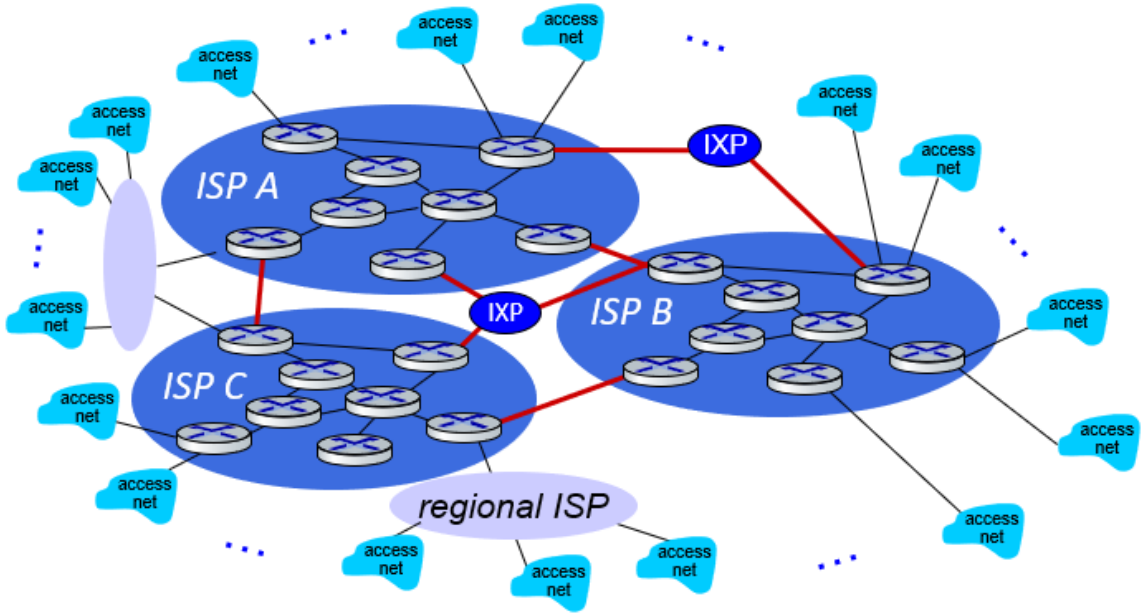
Internet Structure: A Network of Networks

- But if one global ISP is viable business, there will be *competitors*…
  who will want to be *connected*



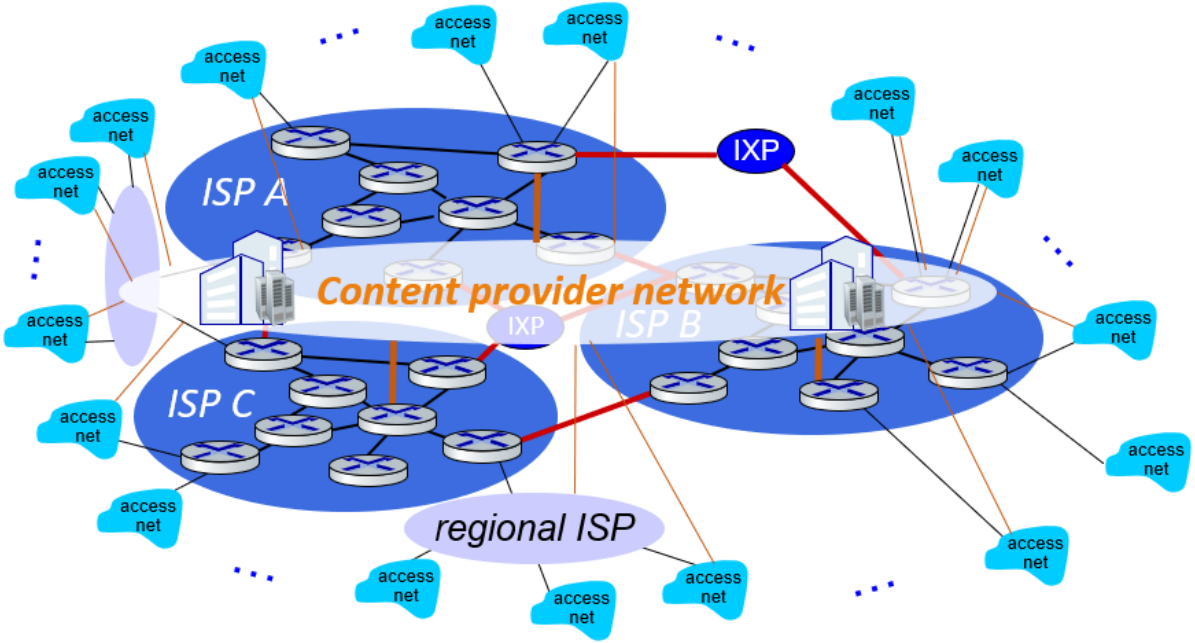
Internet Structure: A Network of Networks

- …and *regional networks* may arise to connect access nets to ISPs…



Internet Structure: A Network of Networks

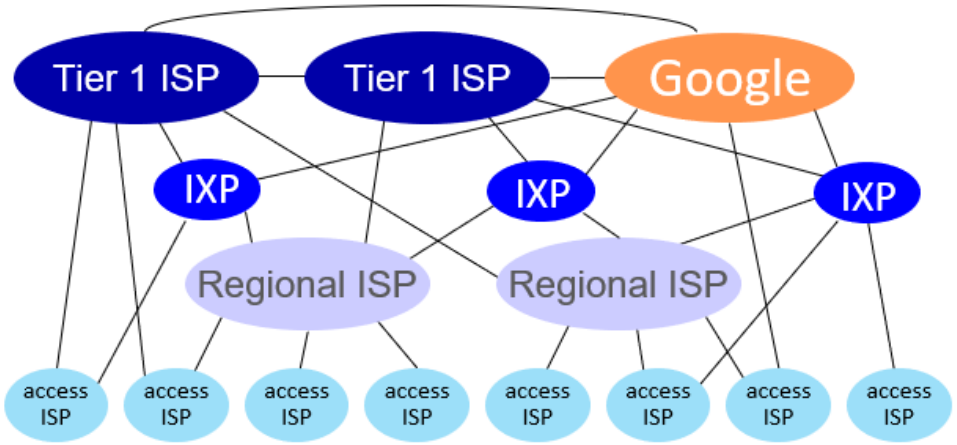
- …and *content provider networks* (e.g., Google, Microsoft, Akamai) may run their own network, to bring services and content close to end users



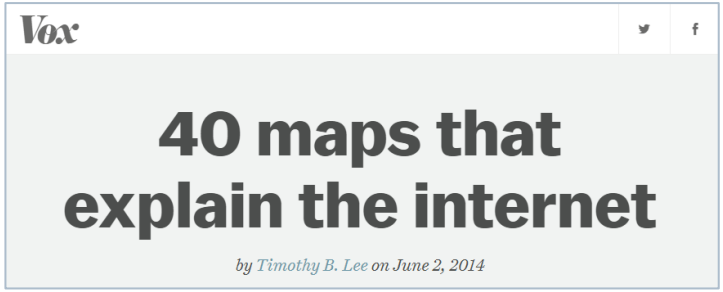
Hierarchical Structure

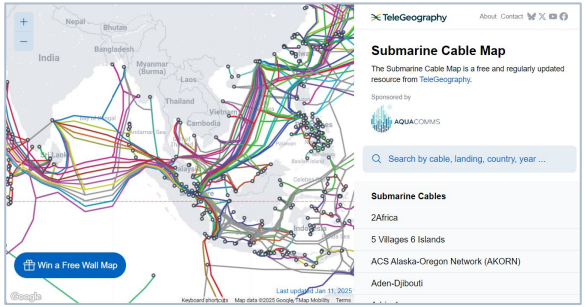
At “center”: small number of well-connected large networks

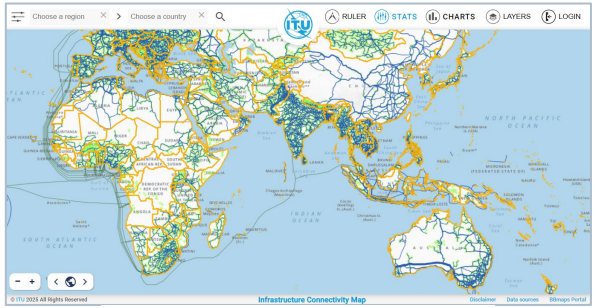
- “Tier-1” commercial ISPs (e.g., AT&T, Verizon, NTT)
  - Have national and international coverage, settlement-free peering
- Content provider networks (e.g., Google, Facebook)
  - Private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs



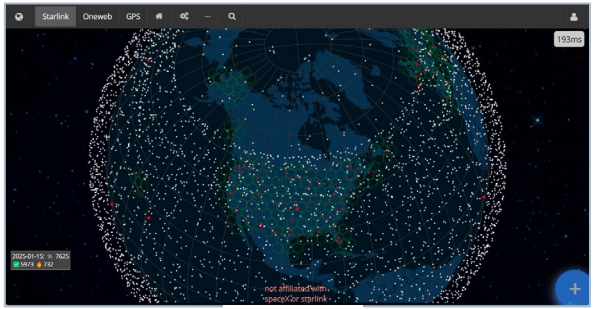
Extra Readings (and Maps for Fun)







ITU - Connectivity map
(submarine and fiber optics)



Satellite